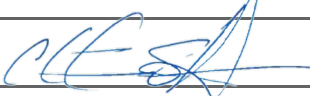


	LOCK OUT SHEET
	NUMBER :
	DATE : JUNE 14TH, 2021
	VERSION : 1

Lock-out STOPEMASTER drill

OWNER :	Opération
UNIT NUMBER	
SECTOR:	Stopemaster Drill

VALIDATED BY STÉPHANE TREMBLAY	
APPROVED BY CHRISTIAN ST-AMOUR	

THIS WORKING INSTRUCTION APPLIES TO :	SECTIONS TO READ :
Every worker around the drill during the application of the instruction	ALL

1. OBJECTIVES

This LOCK OUT SHEET presents the LOCK-OUT method for underground Stopemaster drill.

Here is the list of tasks which need a multiple lock-out **on the main panel master switch AND the diesel engine master switch**

- Work on the hydraulic system
- Slide adjustment
- Changing JAW or cylinder
- Changing selenoïd
- Any work on the electrical system
- Any work involving water and air pressure (filters, cooler, valves, air or water hose, water pump)
- Work on the drifter (ex. Changing shank or nose seal)

Tasks which need a multiple lock out **on the both master switch and gravitational energy**

Tasks that involve the gravitational energy are managed in the last part of this lock out sheet. The two pieces of equipment that requires blocking or put in stable and non-moving position are the boom, the carrier and the drifter

IF A TASK DOESN'T APPEAR ON THIS LIST, A RISK ANALYSIS SHOULD BE DONE

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Lock-out STOPEMASTER drill

2. TRAINING AND MATERIEL

PPE MANDATORY	PERMIT MANDATORY	TRAINING MANDATORY
Site PPE	Explosives permit (Quebec)	- Common core (Ontario) or FMTM (Quebec) - Lock-out training (site or MRI) (If applicable)

NECESSARY MATERIEL	
EQUIPMENT	PRODUCTS
- Multi lock - Sealed, personnel padlock, service padlock - Lock-out/tag-out box (drill) - Mechanical lock - Sealed - Control label	

NO WORKER CAN OVERPASS THE YELLOW RIBBON DURING THE LOCK-OUT PROCEDURE IF HIS PADLOCK IS NOT ON THE MULTI-LOCK OR THE LOCK-OUT BOX

Lock-out STOPEMASTER drill

3. WORK INSTRUCTION

#	ILLUSTRATIONS	PROCEDURE	ADVICE AND RISKS MANAGEMENT
		MULTIPLE LOCK-OUT	
1A		Close and purge the air and water lines on the console and DISCONNECT to air and water hose from the headers when work on air or water system is needed	A small leak on a valve can provoke an accumulation in the piping
1		Press the stop button on the console	
2		Press the emergency stop	
3		Put the master switch on OFF position and lock the <u>main panel</u> master switch with a serial padlock	Always use your left hand to open or close the master switch and place your body on the right side of the panel

Lock-out STOPEMASTER drill

#	ILLUSTRATIONS	PROCEDURE	ADVICE AND RISKS MANAGEMENT
4		Put the master switch on OFF position and lock the <u>diesel engine</u> master switch with a serial padlock	
5		Put the serial padlock key inside the lock-out box Lock the box with a multi-lock and the personals padlocks	
6		Do the start-up test on the console and on the master panel of the Stopemaster	

Lock-out STOPEMASTER drill

#	ILLUSTRATIONS	PROCEDURE	ADVICE AND RISKS MANAGEMENT
7		Do the start-up test on the diesel engine	
		GRAVITATIONAL ENERGY	
2		If you can move the equipment, release the hydraulic system and put the rods or the bit down on the bench	Gravitational energy
3		If you can move the equipment, put the boom on 90 degrees position with the dump cylinder fully extended, in this way you have no pressure in the hydraulic hose of the dump cylinder	
4		If you CANNOT move the equipment, you will have to purge the feed hydraulic cylinder to be able to sit the drifter on the bottom or the top of the boom depending if you drill upper or down hole and drain the dump cylinder to be fully extend	Control and recover the hydraulic oil Don't stay in the moving line of the drifter

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Lock-out STOPEMASTER drill

#	ILLUSTRATIONS	PROCEDURE	ADVICE AND RISKS MANAGEMENT
		If you need to work under the carrier or on the wheel, you need to lift the drill on certified stand, not only lifted with the hydraulic jacks	
		UNLOCKOUT	
12		Do the reverse process	Always use your left hand to open or close the master switch and place your body on the right of the panel